ELAKIYAA — CLIENT & WORKFORCE MANAGEMENT SYSTEM

Complete Technical Specification & Project Architecture Documentation

REACT 19.3 VITE 8 SUPABASE POSTGRESQL TAILWIND CSS 3.4

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1. Project Overview & Business Model

The **ELAKIYAA Client & Workforce Management System** is a cloud-based software platform engineered for **ELAKIYAA**, an enterprise **Manpower Supply & Staffing Agency**.

> 💡 **Real-World Business Model:**
> • **ELAKIYAA (The Company)**: A manpower agency that recruits, manages, and supplies trained field attendants, security staff, and cashiers.
> • **Clients (Petrol Bunks & Enterprises)**: Indian Oil (IOCL), Bharat Petroleum (BPCL), Hindustan Petroleum (HPCL), Shell, Nayara Energy petrol bunks that hire workforce from ELAKIYAA.
> • **Staff (The Workforce)**: Employees deployed at client sites who perform daily shifts and record attendance.
> • **Managers**: Area supervisors from ELAKIYAA who inspect deployed staff and coordinate daily shift rosters.

Primary Problem Solved: Eliminates ghost attendance, manual paper registers, untracked duty allocations, and billing disputes between the staffing agency and client petrol stations.

2. System Architecture & Tech Stack

The project follows a **Modern Serverless Full Stack SaaS Architecture**:

Architecture Layer	Technology Used	Purpose
Frontend Framework	React 19 (Functional Components + Hooks)	High-performance reactive user interface with state isolation.
Build System	Vite 8.3.2	Ultra-fast HMR and optimized production bundle packaging (~2s builds).
Styling & Design	Tailwind CSS 3.4	Utility-first mobile responsive interface with consistent brand theming.
Cloud Backend & Database	Supabase (PostgreSQL Cloud)	Relational database storage, schema definitions, and RLS security policies.
Hosting & CI/CD	Vercel Global Edge Network	Automated build and continuous deployment connected to GitHub `main` branch.
Hardware APIs	WebRTC MediaStream + HTML5 Canvas	Live camera facial capture with automated GPS coordinate & time watermarking.

3. Role-Based Access Control (4 Distinct Portals)

The system provides secure, role-restricted portals for 4 classes of users:

👑 1. Admin Portal (`/admin`)

Complete organizational oversight: User account governance, multi-staff task allocations, attendance overrides with audit logging, salary generation, and system shift configuration.

👔 2. Manager Portal (`/manager`)

Supervises assigned staff teams, monitors client bunk operations, reviews and approves leave applications, and coordinates daily shift coverage.

👷 3. Staff Portal (`/staff`)

Biometric camera punch-in/punch-out with live GPS stamp, task status progress updates, leave submission, and downloadable salary vouchers.

🏢 4. Client Portal (`/client`)

Client bunk dealers monitor deployed staff on their site, confirm daily duty attendance ("Duty Confirm"), initiate direct phone calls, and download monthly invoices/contracts.

4. Core System Modules & Features

1. Tamper-Proof Photographic Attendance:

Staff capture their photo using their front device camera. The system reads real-time GPS coordinates and stamps Employee Name, Timestamp (HH:MM:SS), and City/Address directly onto the canvas. Punch-ins after 10:10 AM automatically calculate exact late penalty minutes.

2. Multi-Staff & Multi-Client Task Delegation:

Admins can select multiple staff members and multiple client bunks simultaneously in one assignment form. Progress tracks seamlessly from `Pending` ➔ `In Progress` ➔ `Completed`.

3. Immutable Audit Logging:

All sensitive operations (attendance edits, leave approvals, salary updates, account deactivations) append an immutable record into the audit log containing User, Action, Timestamp, and IP Address.

4. Client Duty Confirmation:

Petrol bunk dealers log in and click "Confirm Duty" for workers present at their station. This provides legal proof of attendance for monthly billing reconciliation.

5. Database Schema & Data Models

Located in `supabase/schema.sql`, the PostgreSQL relational schema includes:

```
-- 1. USERS TABLE
CREATE TABLE users (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    name VARCHAR(100) NOT NULL,
    mobile VARCHAR(15) UNIQUE NOT NULL,
    password VARCHAR(255) NOT NULL,
    role VARCHAR(20) CHECK (role IN ('admin', 'manager', 'staff', 'client')),
    photo TEXT,
    status VARCHAR(20) DEFAULT 'active',
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- 2. ATTENDANCE TABLE (With Photographic & Geo Evidence)
CREATE TABLE attendance (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID REFERENCES users(id),
    date DATE NOT NULL,
    punch_in TIME,
    punch_out TIME,
    status VARCHAR(20) CHECK (status IN ('present', 'absent', 'late')),
    late_minutes INT DEFAULT 0,
    photo_url TEXT,
    location_address TEXT,
    coordinates POINT
);

-- 3. TASKS TABLE (Multi-Entity Allocation)
CREATE TABLE tasks (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    title VARCHAR(255) NOT NULL,
    due_date DATE NOT NULL,
    priority VARCHAR(20) CHECK (priority IN ('high', 'medium', 'low')),
    status VARCHAR(20) CHECK (status IN ('pending', 'in_progress', 'completed')),
    remarks TEXT
);

-- 4. AUDIT LOGS (Immutable Activity Log)
CREATE TABLE audit_logs (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_name VARCHAR(100),
    action VARCHAR(50),
    module VARCHAR(50),
    description TEXT,
    ip_address VARCHAR(45),
```

```
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

6. Evaluation & Demo Credentials

Role	Sample User	Mobile Number	Password	Primary Portal
👑 **System Admin**	Arjun Sharma	`9000000001`	`admin123`	`/admin`
👔 **Operations Manager**	Priya Nair	`9000000002`	`manager123`	`/manager`
👷 **Field Staff**	Anitha Devi	`9000000004`	`staff123`	`/staff`
🏢 **Client (Bunk Dealer)**	Indian Oil Petrol Bunk	`9100000001`	`client123`	`/client`

> ⚡ **Evaluator Tip:** On the live login page at https://client-management-system-iota.vercel.app, click any of the 4 quick test pills to instantly log in without typing credentials!

7. GitHub Code Navigation Guide

When reviewing the GitHub repository (`https://github.com/indhurahul1108-lgtm/client-management-system`), use this map to inspect files:

Directory / File	Contents & Code Location
`frontend/src/pages/admin/`	**13 Admin Pages**: Dashboard, Assign Tasks, Manage Staff, Attendance, Salary, Audit Log, User Management, System Settings.
`frontend/src/pages/manager/`	**10 Manager Pages**: Staff Roster, Client Oversight, Attendance Review, Leave Approval, Team Tasks.
`frontend/src/pages/staff/`	**Staff Pages**: Camera Punch In/Out (`Attendance.jsx`), Work Updates (`StaffWork.jsx`), Payslips, Leaves.
`frontend/src/pages/client/`	**Client Pages**: Duty Confirmation (`ClientDashboard.jsx`), Work Monitoring, Document Locker, Service Requests.
`frontend/src/context/`	`AuthContext.jsx` : Session management, role-based route protection, login/logout functions.
`supabase/`	`schema.sql` & `seed.sql` : PostgreSQL relational database tables, foreign keys, and seed records.
`README.md` & `DOCUMENTATION.md`	Full system documentation viewable directly on GitHub repository homepage.